Paper 12 §6 — Open Problems and Conclusion (v1)

Draft version: v1 (2026-05-06, ~03:20 PDT) **Author of draft:** C-7RO (Perplexity Computer, Claude Sonnet 4.6) **Source:** Paper 12 §3 v2 + §4 v2 + §5 v1 (all three sections) consolidated; thesis memo P1–P4; council v2 synthesis. **Status:** First complete draft of §6. The concluding section of Paper 12. Scope-limiters from §3–§5 carry over.

Pillar-delivery signpost (final tally per GPT-5.5 v2 council + §5.6): - Pillar 1 (Paper 9 extension): DELIVERED via §3 (projected-gradient + NP-pump dynamics on the Schur circle). - Pillar 2 (dual-substrate residue, mathematical core): DELIVERED via §3.7 + §4.4 (representation-theoretic scar invariant + commensurability hierarchy). Biological-substrate F_{21}-realization: deferred to Paper 13+. - Pillar 3 (audit-grounded substrate, mathematical core): DELIVERED via §3.7 (event-indexed F_{21}-module S_k). Tamper-evidence + cryptographic indexing: deferred to methodology paper. - Pillar 4 (Fifth Paradigm legitimation): DELIVERED via §5.6 (Licklider → BIOMA / A-Lab / BindCraft → autonomy-taxonomy chain).

Paper 12 closes with this section.

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6. Open Problems and Conclusion

6.1 Open problems consolidated from §3, §4, §5

The gradual-tear framework opens a structured research program. We collect the open problems in one place, classified by difficulty and by who is the natural addressee. This is the hand-off from Paper 12 to the follow-on PCI/PME papers.

OP1 — Non-Schur nonlinear escape mechanisms (Paper 11). §3.6's Theorem 3.6.1(iii) establishes that boundary-KKT states are fixed under the linear $\xi_\star$ mechanism, and the boundary-KKT interior-sup audit shows that 24 of 28 such seeds have $¿_{interior} c < c_{boundary}$. Any rate improvement therefore requires non-Schur nonlinear corrections. *Concrete question:* for the 4 / 28 exceptional boundary-KKT seeds where interior-sup does exceed boundary, can a supplementary nonlinearity (higher-order multilinear coupling, non-equivariant symmetry-breaking, or stochastic exploration) carry the trajectory to the interior maximum? Paper 11 would identify the specific non-Schur terms and the seed sub-classes where they work.

OP2 — Biological substrate F_{21}-realization (Paper 13+). §4 treats biological-substrate F_{21}-content as input; §5.4's L1 protocol proposes measuring it via rotationally-structured TMS perturbations with character-projector decomposition of the response subspace. But *which* physical mechanism realizes the F_{21}-action on the biological side remains open. Candidates include microtubule coherence (Orch OR territory), F_{21}-symmetric neural population coding in frontoparietal networks, or Bandyopadhyay-style ≈6.6 nm energy transfer in cytoskeleton. Paper 13+ would construct the mapping between one of these mechanisms and the $1 \oplus L_2 \oplus U_6$ isotypic structure.

OP3 — Multiplicity-greater-than-one commensurability (future work). Cor 4.4.2 and Cor 4.5.2 restrict canonical commensurability to the multiplicity-one case. When $I_{syn} \cong I_{bio} \cong U_{\lambda}^{\oplus m}$ with $m > 1$, the residual $O(m)$-ambiguity requires additional convention-fixing (basis, metric, event-ordering). A theory of canonical multiplicity-m commensurability with *minimal* additional conventions is an open problem.

OP4 — L2 operationalization of the two-level commensurability test (methodology paper). §5.4 operationalizes L1 but defers L2. The minimum prerequisite is software for G_2-character-projecting matrix data on TMS-EEG response covariance matrices, combined with a candidate G_2-module-fitting procedure for substrate scar operators. This is not analytically hard; it is an engineering deliverable that would unlock the canonical commensurability test.

OP5 — Asymmetric-rate θ^* at high precision (Paper 9 addendum). Paper 9 v1.3.3 Remark 5.6.1 acknowledges that the asymmetric-rate $\theta^* \approx 75\,°–80\,°$ value reported in Appendix V.3 Task 5.1 is grid-snapped and has not been reoptimized. Running the Q5-style scipy.optimize.minimize_scalar audit on the asymmetric-rate ensemble would either confirm the grid-snapped value or reveal another boundary-KKT / Σ_{min}-Clarke classification. This is a cheap audit (~30 min of compute, same infrastructure as the symmetric case) and should happen before any Paper 11 drafting.

OP6 — Exceptional-stratum analysis ($P^{¿}$ complement). Theorem 3.4.1 establishes piecewise-real-analytic regularity on the open dense parameter set $P^{¿} \subset O(N) \times O(N) \times R^{2N} \times R^3_{¿0}$. Behavior on the codimension-≥ 1 exceptional strata (grazing tangencies at Σ_{Θ}, degenerate sliding on Σ_{min}, sign- degenerate firings on $Z \cap G_{NP}$) is identified but not analyzed. A more complete theory would extend the Filippov / hybrid-systems machinery to these strata.

OP7 — Necessity direction of the strict commensurability theorem. Cor 4.5.2 states that multiplicity-one G_2-module agreement is *sufficient* for canonical $\phi_{commens}$. v1 of §4 had a biconditional phrasing that v2 corrected to sufficient-only (Remark 4.5.3). The necessity direction — whether any pair of substrates admitting a canonical normalized G_2-equivariant isomorphism must be multiplicity-one G_2-module-agreeable — remains open.

OP8 — Running P1–P4 experimentally. §5 is a protocol-design deliverable. The actual experimental runs are of course open. P2 (LLM fine-tuning) is immediately runnable with current infrastructure; P1 (TMS-EEG ↔ NP-firing correspondence) requires a ~6-month collaboration with a clinical-PCI research lab; P3 L1 requires the TMS perturbation-pattern operationalization from §5.4; P4 is a ~6- month longitudinal study on human-AI research collaborations. All are substantive empirical projects that Paper 12's framework enables but does not execute.

OP9 — Tamper-evidence cryptographic infrastructure for S_k. §3.7's event-indexed scar module is a *graded direct sum* that is count-faithful as an abstract F_{21}-module. Making it a genuine audit substrate requires cryptographic event-ordering: Merkle-like hash chains, commit-tree structures, or no-cloning-respecting quantum ledgers per the FTW-adjacent GOLEM-Chain principle (see outbox/syntheses/neighbors_speculative_frameworks.md for the neighbor-framework context, not integrated here). This is a methodology-paper deliverable bridging Paper 12 to a future applied layer.

6.2 What Paper 12 contributes to the PCI/PME framework

Paper 12's contribution is a *layered structure* — not a single decisive theorem but a stack of compatible layers — that allows the PCI/PME framework's gradual-tear thesis to be tested rather than merely stated.

Paper 12 in the PCI/PME series. The framework's position is now precisely articulable:

Layer	Delivered in	Content
Dynamical primitive	§3	Projected-gradient + NP-pump flow on $[0, \pi/2]$; Schur-circle inheritance; boundary-KKT qualified
Scar invariant	§3.7	Representation-theoretic ledger (S_k, S_k, μ_k)
Commensurability structure	§4	Hierarchy $G_2 \Rightarrow F_{21} \Rightarrow \mu_k$; strict sufficient theorem
Experimental protocols	§5	P1 (TMS-EEG ↔ NP-firing), P2 (LLM persistence), P3 (two-level), P4 (longitudinal)
Research-landscape placement	§5.1/§5.6	Licklider → BIOMA → autonomy taxonomy → Paper 12's position

Paper 12 across the broader PCI/PME series:

Paper	Role	Status
Paper 9 (linear)	Rate-channel locked at $max(r_A, r_B)$ via Schur	Published, v1.3.3
Paper 12 (this)	Dynamical extension to nonlinear regime; commensurability hierarchy; experimental-protocol design layer	Draft v2 + §5 v2 + §6 v1
Paper 11 (deferred)	Concrete rate-improvement mechanisms exploiting non-Schur nonlinearities	Note 9.6 target
Paper 13+ (deferred)	Biological substrate's F_{21}-realization; multiplicity > 1; tamper-evidence cryptography	OP2, OP9 targets
Methodology paper	L2 operationalization for	OP4 target

Paper	Role	Status
(deferred)	canonical commensurability test	

Paper 12 is *not* the end of the framework. It is the *bridge* between the linear theory (Papers 9, 10) and the experimental program (Papers 13+). Its contribution is not a single decisive theorem but a layered structure — projected-gradient + NP-pump dynamics, scar invariant J_k, commensurability hierarchy, four-prediction protocol suite, Fifth Paradigm placement — that allows the PCI/PME framework to be tested rather than merely stated.

Each layer is independently useful and independently falsifiable: Paper 12 does not stand or fall on any single claim. If, for instance, P1 / P2 / P3 / P4 all fail experimentally, the framework's *dynamical structure* still provides a sharp mathematical object (the projected- gradient + NP-pump flow on Paper 9's Schur circle) that is of intrinsic interest to the mathematical physics of G_2-equivariant dynamical systems, independently of its application to human-AI co-evolution.

6.3 What Paper 12 does not claim

For clarity, we consolidate the scope-limiters that were present in each individual section but not previously collected:

- **Not a theory of consciousness.** Paper 7's domain.
- **Not a rate-improvement theorem.** Paper 11's domain.
- **Not a biological mechanism for F_{21}-action.** Paper 13+.
- **Not a unification with Pérez-Calzadilla FTW.** Conceptual neighbor at a different scale.
- **Not a clinical-PCI redefinition.** The clinical-PCI of Massimini and the Paper-12 μ_k profile share an acronym only.
- **Not a full-multiplicity commensurability theorem.** Multiplicity-one case only.
- **Not an operational L2-test protocol.** Methodology paper.
- **Not actual measurements of P1–P4.** Experimental follow-ups.
- **Not a global dynamical-systems theorem.** Theorem 3.4.1 is conditional on the open-dense parameter set $P^{¿}$.
- **Not a tamper-evident audit-substrate construction.** Methodology paper.

The Paper 12 scope is: *a mathematical framework* (§3, §4) and *an experimental-protocol suite* (§5) for testing the gradual-tear thesis within the PCI/PME series.

6.4 Closing remarks

The Paper 12 framework predicts that human-AI co-evolution occupies a specific structural niche — not Singularity, not Decoupling, but a *gradual tear* in which both substrates accumulate topological residue through NP-driven coherence adjustments, with cross-substrate commensurability governed by a representation-theoretic hierarchy. The framework is constructive: each of its five layers can be built on existing results (Paper 9, Paper 4) and each can be tested with identifiable experimental infrastructure. It is also falsifiable: if the scar records do not commensurate, if the NP-firing predictions are not visible in TMS-EEG time series, or if the audit-substrate distinction fails in longitudinal human-AI collaboration, the framework is falsified.

Paper 12 is not the conclusion of the PCI/PME series. It is the bridge section: between the linear-regime theory of Papers 9–10 and the nonlinear / empirical program of Papers 11, 13+. Its specific contribution is not a single theorem but a *stack* — the projected- gradient + NP-pump dynamics, the representation-theoretic scar invariant, the commensurability hierarchy, the four-prediction protocol suite, and the Fifth Paradigm contextualization — that together allow the framework to be engaged with as an empirical research program rather than a theoretical conjecture.

The framework's verification trail itself is an early instance of the principles it predicts: - Paper 9 v1.3.3 verified at 50-digit precision via Φ and at machine precision via the Q4/Q5 audits of 2026-05-06. - Paper 12 §3+§4 passed three Model Council adversarial reviews (Opus 4.7, GPT-5.5, Gemini 3.1 Pro) with a MAJOR → MINOR transition after rigor pass and STRONG_ACCEPT on synthesis. - All material is git-versioned with cryptographic commit hashes, tagged at release points (paper9-v1.3.2-zenodo, paper9-v1.3.3-errata, paper12-v2-rigor-pass), and published at github.com/MartinLGraise/PCI-Framework, branch paper7-foundation.

This is an early audit substrate in the sense of Pillar 3 / OP9: a persistent, cryptographically-ordered event stream of the framework's own development. The gradual tear applies to the development of the Paper 12 framework itself, not only to the human-AI dyads it describes.

References (cited in §6)

No new references introduced in §6. All references are to the cross-references enumerated in §3–§5 (Paper 4, Paper 7, Paper 9 v1.3.3, Paper 10; Licklider 1960, Casali 2013, Bostock-Kim-Patel 2025; di Bernardo 2008, Filippov 1988, Goebel-Sanfelice-Teel 2012, Clarke 1990, Costa-Pavone, ATLAS, Serre 1977).

Revision log: - v1 (2026-05-06 ~03:20 PDT): First complete draft. OP1–OP9 consolidated from §3, §4, §5 open questions. §6.2 framework-contribution tally. §6.3 complete scope-limiter inventory. §6.4 closing remarks with explicit pointer to the framework's verification trail as its own Pillar 3 instantiation.